

## **place.tcl**

```
# Description : Placement script – routing layers, LVT usage, tie cells,  
#             placement options, congestion-aware flow, and reports.  
# Tool       : Synopsys ICC2 (or similar implementation tool)  
# Requirements :  
# - tech_setup.tcl, user_setup.tcl  
# - ${design}_pre_place_done block already created  
#  
# Key Variables (set in user_setup.tcl or runset):  
# max_core          - Max CPU cores  
# ndm_lib_name      - Library name under db/  
# design            - Top-level design name  
# min_route_layer   - Minimum routing layer to use  
# max_route_layer   - Maximum routing layer to use  
# allow_lvt_place    - If 0, LVT cells are disabled for placement  
# enable_ccd_place   - Enable CCD during placement (0/1)  
# cong_aware_placement - Enable congestion-aware placement flow  
#                   (0/1)  
# place_two_pass_flow - Enable two-pass place_opt refinement (0/1)  
# setup_uncertainty_place - Percent of clock period used as setup  
#                   uncertainty  
# do_place_qor_check - If 1, dump QoR reports and maps  
# stop_after_place   - If 1, stop after placement stage  
  
# Read Technology & User Settings  
source -e -v ./scripts/tech_setup.tcl  
source -e -v ./scripts/user_setup.tcl
```

## # Multi-CPU Setting

```
set_host_options -max_cores $max_core
```

## # Open NDM Library and Create Placement Block

```
if { [file exists "db/${ndm_lib_name}"] } {  
    puts "\n[INFO] Library 'db/${ndm_lib_name}' exists, opening...\n"  
    open_lib db/${ndm_lib_name}  
    copy_block -from ${design}_pre_place_done -to ${design}_place_done  
    current_block ${design}_place_done  
    link  
} else {  
    puts "\n[ERROR] NDM Library 'db/${ndm_lib_name}' does not exist.  
Please check.\n"  
    return  
}
```

## # Routing Layer Settings

```
puts "\n[INFO] Setting routing layer range: $min_route_layer ..  
$max_route_layer\n"  
set_ignored_layers \  
    -min_routing_layer $min_route_layer \  
    -max_routing_layer $max_route_layer
```

## # LVT Usage Control

```
if { ! $allow_lvt_place } {  
    puts "\n[INFO] Disabling LVT cells for placement.\n"  
    set_attribute [get_lib_cells */*LVT*] dont_use true
```

```
set_lib_cell_purpose -include none [get_lib_cells */*LVT*]  
}
```

# High-drive Cells – Example: Don't use \*X32\*

```
puts "\n[INFO] Disabling very high drive cells (*X32*).\n"
```

```
set_attribute [get_lib_cells */*X32*] dont_use true
```